Introduction to Theoretical Computer Science

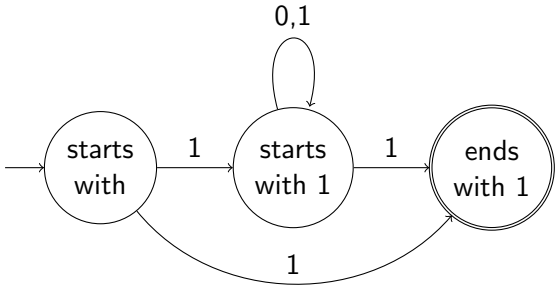
CS2490 - Session 6: Non deterministic Finite Automata

Pierre Cagne

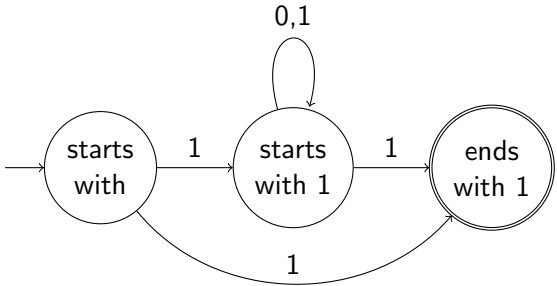
Fall 2026

August 28, 2026

DFA or not?

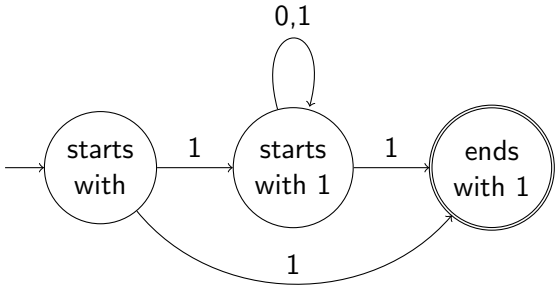


DFA or not?



Two sources of nondeterminism:

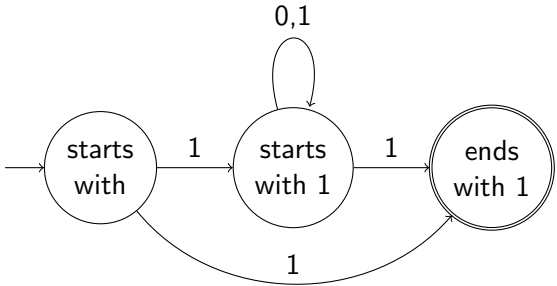
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Two sources of nondeterminism:

- Nowhere to go

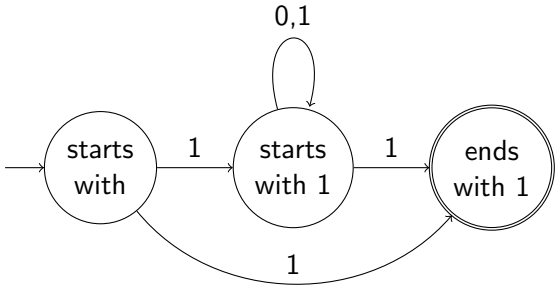
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Two sources of nondeterminism:

- Nowhere to go when reading a 0 in the initial state, and any letter from "ends with 1".

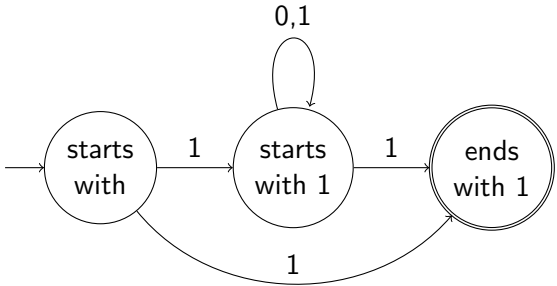
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Two sources of nondeterminism:

- Nowhere to go when reading a 0 in the initial state, and any letter from "ends with 1".
- Too many places to go

DFA or not?



Two sources of nondeterminism:

- Nowhere to go when reading a 0 in the initial state, and any letter from "ends with 1".
- Too many places to go when reading a 1 in the state "init" and "starts with 1".

Definition of NFA

Let Σ be an alphabet. A nondeterministic finite automaton (NFA) $\mathcal{A}$ is given by:

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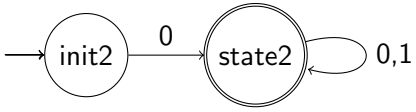
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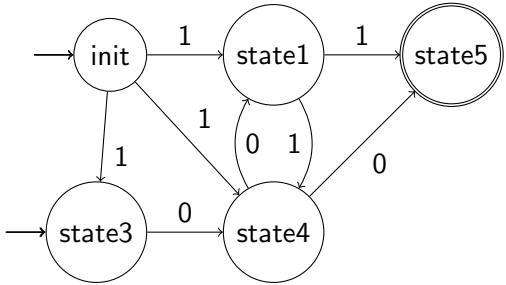
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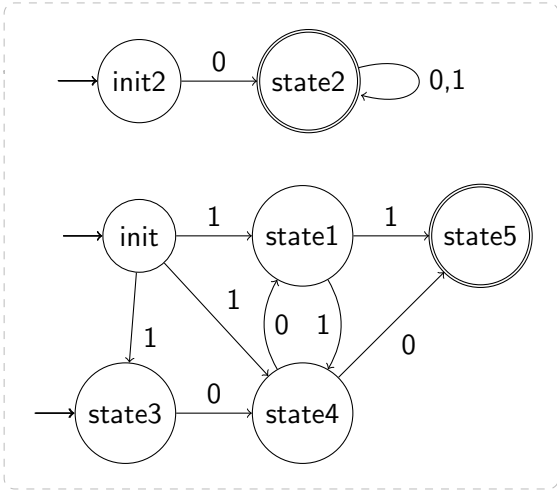
We write $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$.

Example

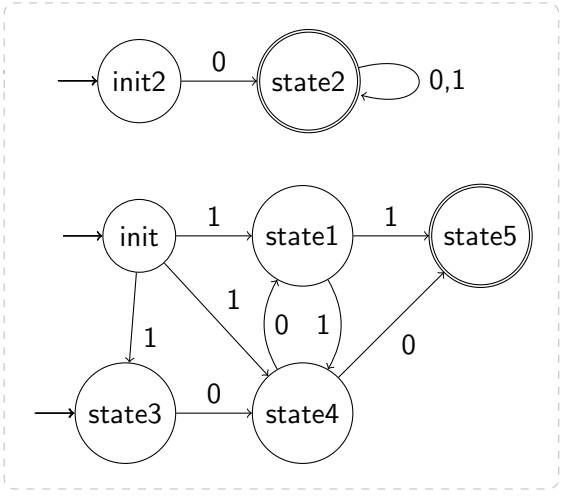

init2
0
state2
0,1


init
1
state1
1
state5
1
1
0
1
0
state3
0
state4

Example

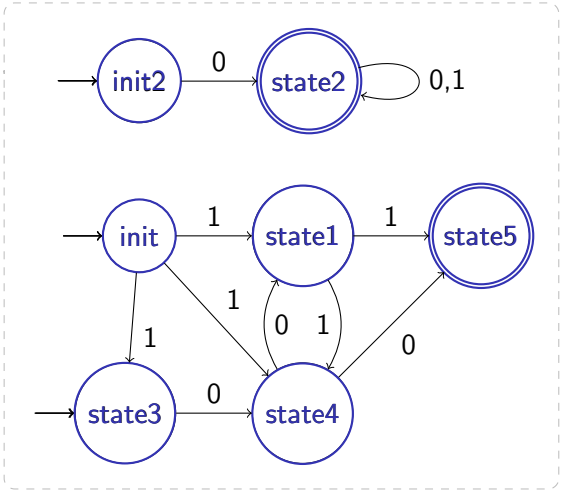

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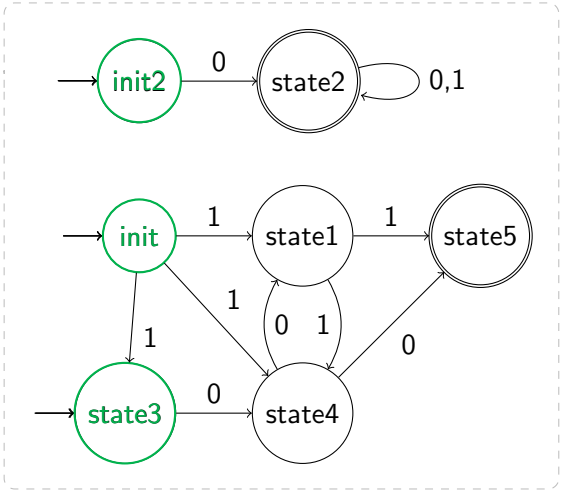
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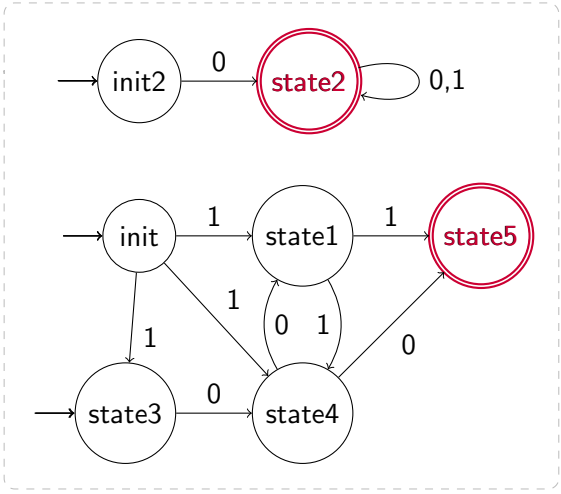


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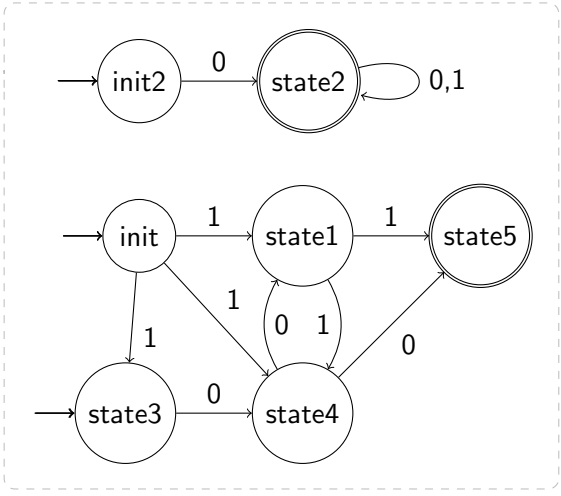
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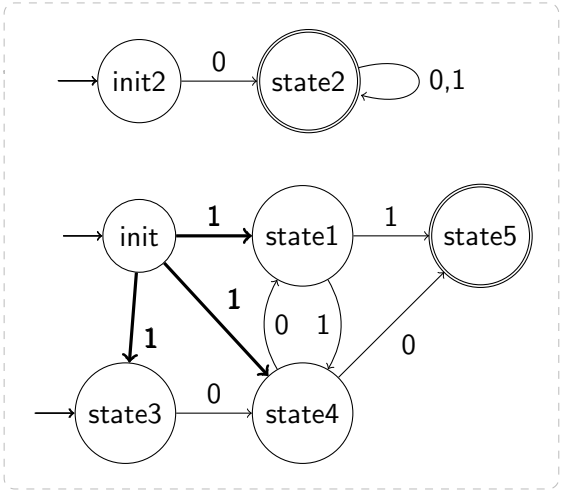
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$\Delta(0, \text{init}) = \emptyset,$

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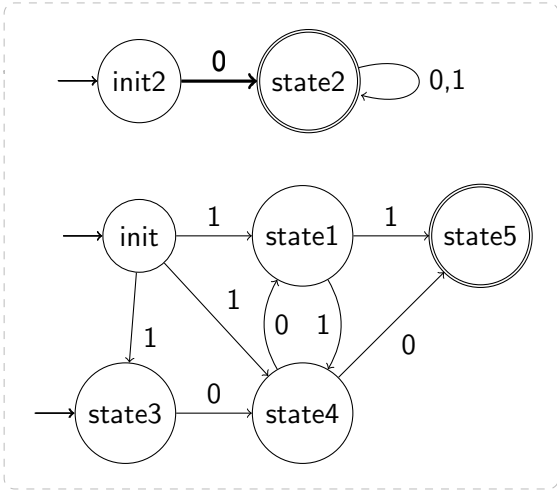
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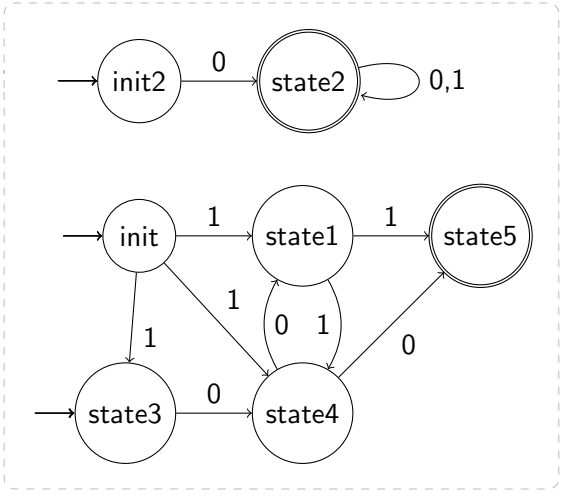
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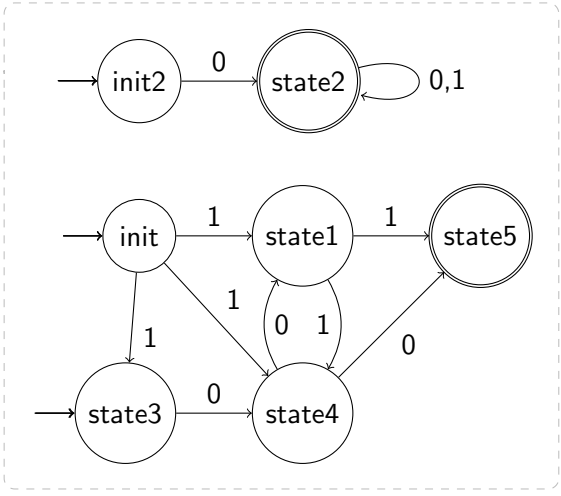
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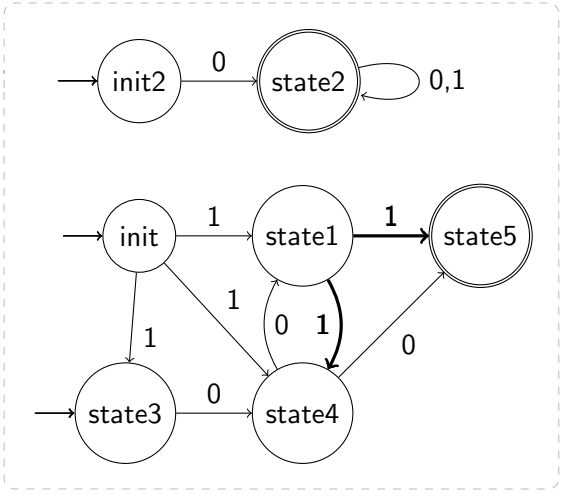
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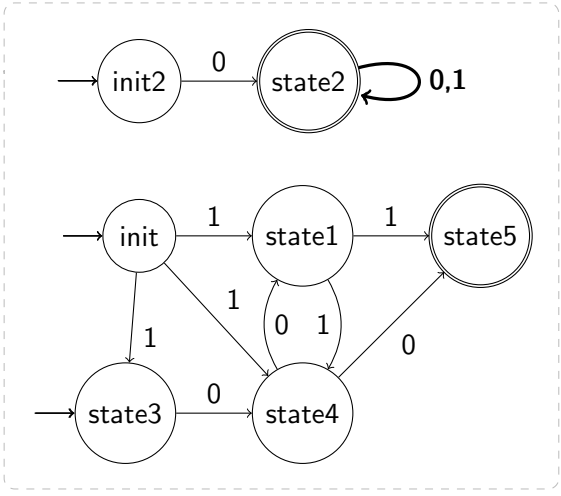
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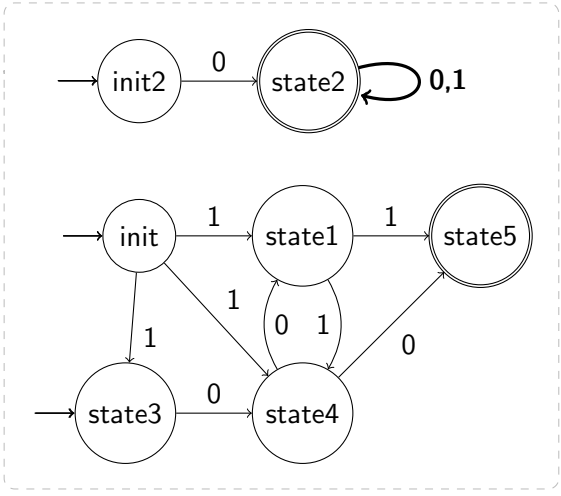
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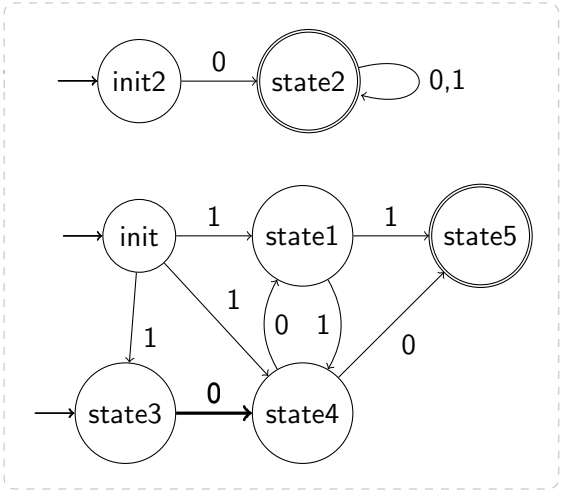
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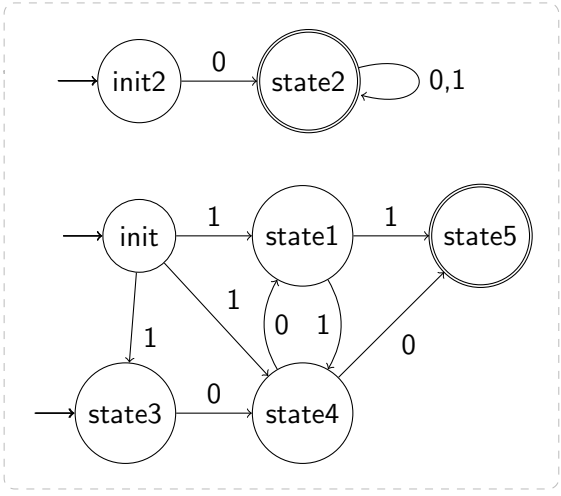
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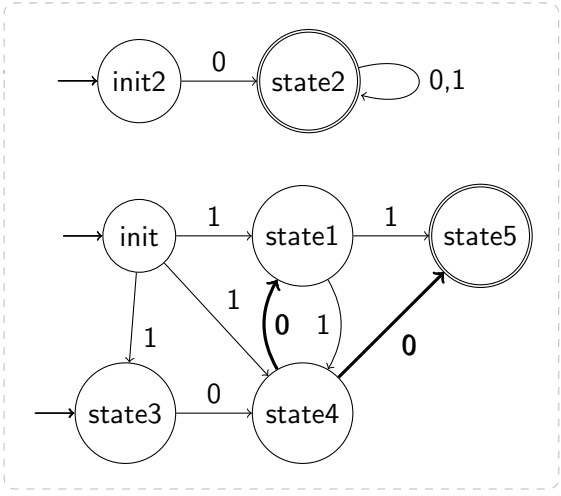
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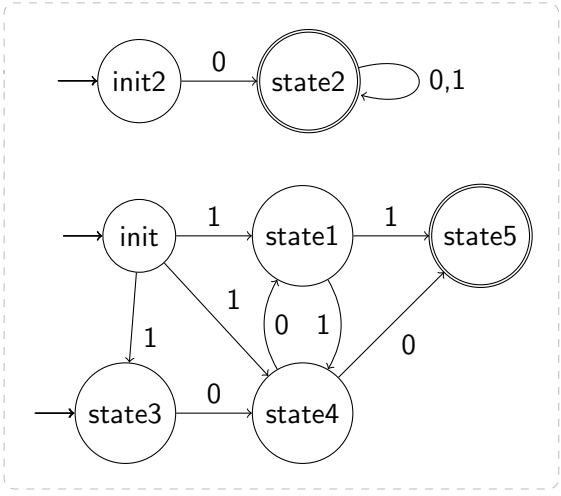
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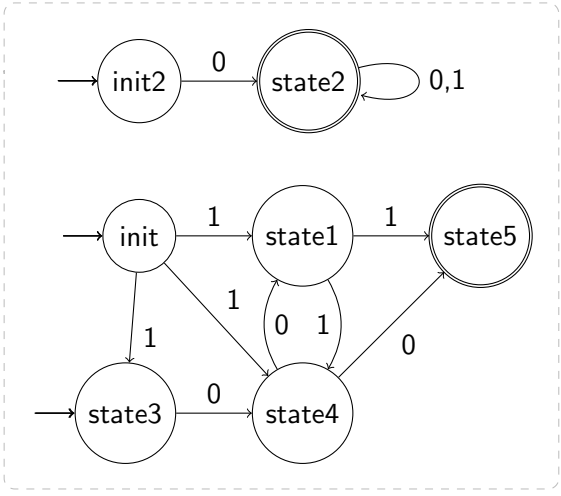
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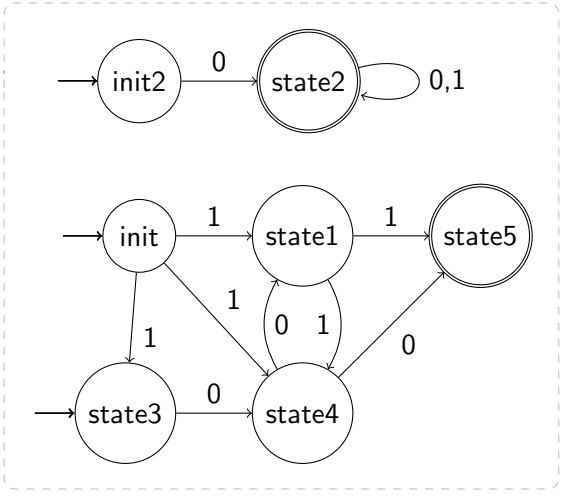
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$$I = \{\text{init}, \text{init2}, \text{state3}\}$$

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$\Delta(0, \text{init}) = \emptyset$, $\Delta(1, \text{init}) = \{\text{state1}, \text{state3}, \text{state4}\}$
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A computation (or run) of $\mathcal{A}$ on a word $w = w_1 w_2 \dots w_n \in \Sigma^*$ is a sequence of states $s_0, s_1, \dots, s_n \in Q$ such that:

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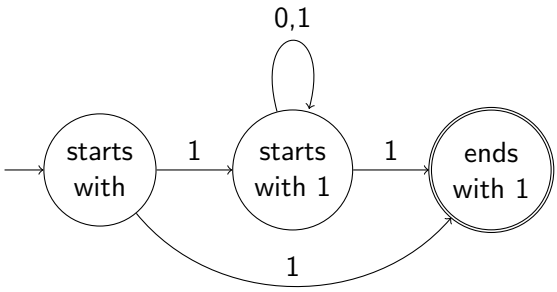
Such a computation is successful when $s_n \in F$.

Computation (cont'd)

Main difference with DFA: computations are far from unique!

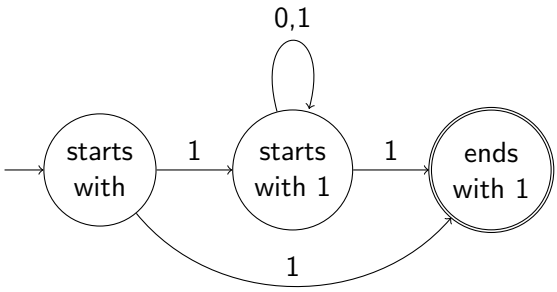
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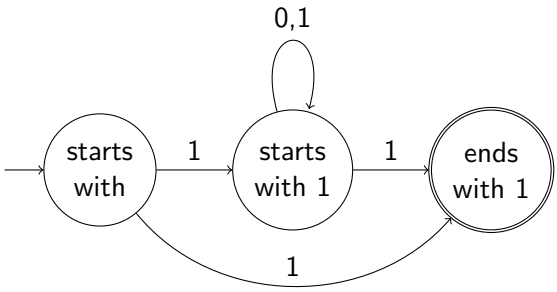
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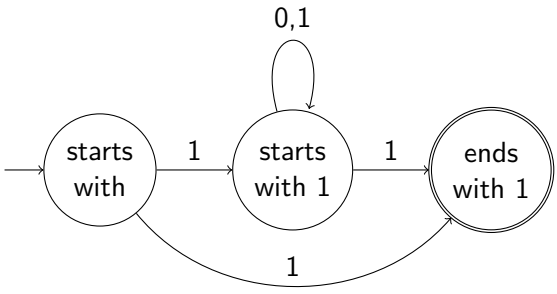


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We then say that $\mathcal{A}$ accepts a word w when there exists **at least one** successful computation of $\mathcal{A}$ on w.

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The language accepted by $\mathcal{A}$ is the set of all words accepted by $\mathcal{A}$:

$$\mathrm{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \exists \text{ a successful computation of } \mathcal{A} \text{ on } w\}$$

Extended transition relation

The extended transition relation $\Delta^* : \Sigma^* \times Q \to \mathcal{P}(Q)$ is defined as follows:

- $\Delta^*(\epsilon, q) = \{q\}$ for all $q \in Q$ (the empty word leads to the same state),
- $\Delta^*(wa, q) = \bigcup_{p \in \Delta^*(w,q)} \Delta(a, p)$ for all $q \in Q$, $a \in \Sigma$, $w \in \Sigma^*$ (reach a bunch of states by reading w, then from each of them read a).

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Restate the definition of the language accepted by $\mathcal{A}$:

$$\mathrm{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \bigcup_{i \in I} \Delta^*(w, i) \cap F \neq \emptyset\}$$

Every DFA is a NFA

Let $\mathcal{A} = (\Sigma, Q, \delta, q_0, F)$ be a DFA. Then it can also be seen as a NFA!

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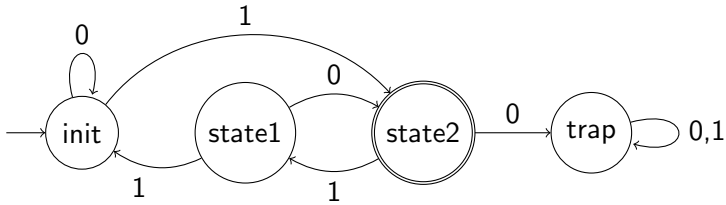
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- Its transition relation Δ is defined by $\Delta(a, q) = \{\delta(a, q)\}$.

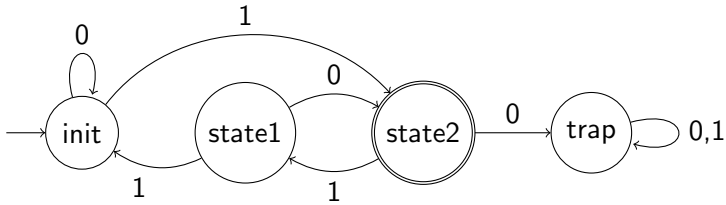
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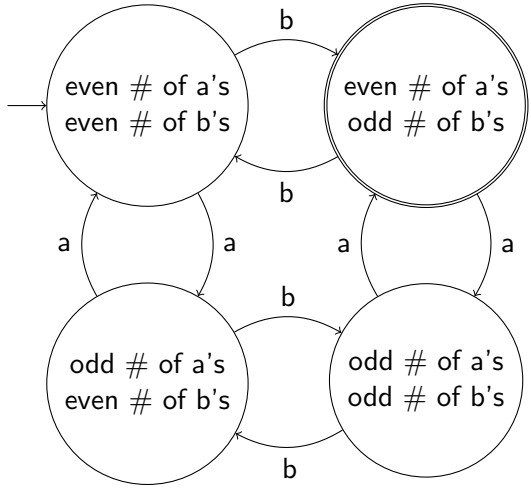
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And its NFA version:



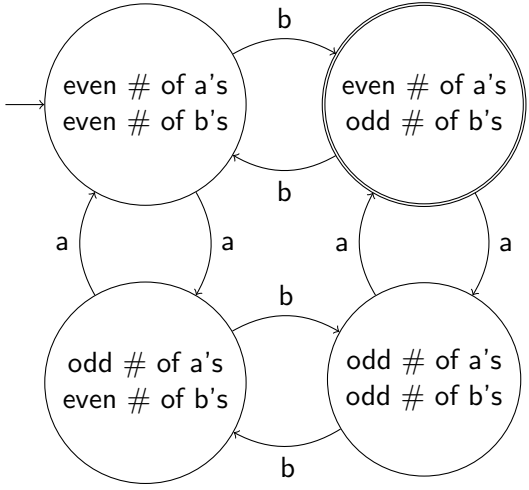
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Next time

How to do the converse?

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I.e., how to transform any NFA into an *equivalent* DFA?